

# Every Morse–Schilling crystal edge is an exchange move, and the exchange move is strictly more general

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*The commit named above contains the mathematics of this note; nothing below this cover block has changed since.*

## Abstract

Morse and Schilling [4] equip the set of two-factor affine factorisations of  $w \in \tilde{S}_n$  with a  $U_q(\mathfrak{sl}_2)$ -crystal, whose raising operator  $\tilde{e}_1$  is defined after fixing a parameter  $x \in [n]$  with  $x \notin \text{content}(u) \cup \text{content}(v)$  and pairing the two contents with respect to  $x$ . In [1] we gave an exchange move on the same objects which uses no such parameter. We settle the relation between the two. Let  $\mathcal{B}(S, T)$  be the set of *all*  $\tilde{e}_1$ -images of  $u_S u_T$ , taken over *every* admissible  $x$  (and, when no  $x$  is admissible, over every instance of the reduction of [4, §3.4]), and let  $\mathcal{E}(S, T)$  be the set of exchange moves of [1, Thm. 4.1]. We prove  $\mathcal{B}(S, T) \subseteq \mathcal{E}(S, T)$  for every length-additive  $(S, T)$  (Theorem 3.2), and we show the containment is strict, the smallest witness being  $n = 5$  (Theorem 4.1). We then characterise the difference exactly (Theorem 4.2): writing the factorisation as a bracket word and letting  $Q$  be its slot-wise low-point profile, membership in  $\mathcal{E}$  is *local* minimality of  $Q$  — the last point of the plateau that opens a run of  $S$  — whereas membership in  $\mathcal{B}$  is *global* minimality of  $Q$  over a full period beginning at an admissible cut. The  $x$ -pairing is therefore not a removable decoration: it is exactly the device that promotes local minimality to global minimality, and it buys canonicity (one well-defined operator, invertible, satisfying the Stembridge axioms) at the price of coverage. The intended reading of Theorem 4.1 as “the  $x$ -free shadow of the whole family of crystal structures” is *false* in the direction one would guess: the union over all  $x$  is strictly smaller, not equal.

## 1 Introduction

### 1.1 The question, and two of my own sentences in tension

Let  $n = k + 1 \geq 2$ , let  $\tilde{S}_n$  be the affine symmetric group, and for a proper subset  $S \subsetneq \mathbb{Z}/n$  let  $u_S$  be the unique cyclically decreasing element with support  $S$  [1, Lem. 2.2]. Call a pair  $(S, T)$  of proper subsets *additive* if  $\ell(u_S u_T) = |S| + |T|$ ; equivalently  $u_S u_T$  is a two-factor affine factorisation in the sense of Lam [2], with  $\text{content}(u) = S$  and  $\text{content}(v) = T$  in the notation of [4].

Morse and Schilling [4, Def. 3.2] define  $\tilde{e}_1$  on such a pair after fixing  $x \in [n]$  with  $u, v \in S_{\hat{x}}$ , i.e.  $x \notin S \cup T$ ; write

$$\mathcal{X}(S, T) = \mathbb{Z}/n \setminus (S \cup T)$$

for the set of admissible parameters. Theorem 4.1 of [1] gives a move on the same objects and fixes no parameter. The question this note answers was posed as: is the exchange move the  $x$ -free shadow of the whole family  $\{\tilde{e}_1^{(x)}\}_{x \in \mathcal{X}}$ , i.e. is

$$\bigcup_{x \in \mathcal{X}(S,T)} \{\tilde{e}_1^{(x)}(uv)\} \setminus \{\mathbf{0}\} \stackrel{?}{=} \{\text{Theorem 4.1 moves on } uv\}. \quad (\text{Q})$$

Two sentences in my own notes of 21 September were in tension, and the first business of this note is to write their quantifiers out.

- (a) “ $\{\tilde{e}_1 \text{ edges}\} \subsetneq \{\text{my moves}\}$ ”, recorded after an exhaustive comparison at a *single*  $x$ .
- (b) “ $|L_1| - |R_1| = |S| - |T|$ , so my hypothesis  $|S| > |T|$  is strictly stronger than Morse–Schilling’s firing condition  $L_1 \neq \emptyset$ ; hence  $\tilde{e}_1$  is never  $\mathbf{0}$  in my regime” — from which I had concluded that  $\tilde{e}_1$  can fire where Theorem 4.1 offers nothing, contradicting (a) read as an unrestricted containment.

Both statements are true and they do not conflict. Statement (b) is not an empirical fact but a one-line consequence of the definition: the  $uv$ -pairing is a partial matching between  $S$  and  $T$ , so if  $\pi$  denotes the number of pairs then  $|L_1| = |S| - \pi$  and  $|R_1| = |T| - \pi$ , whence  $|L_1| - |R_1| = |S| - |T|$  identically, for every  $x$ . The inference drawn from it was the error, and the broken assumption was about *my own* theorem: the hypothesis  $|S| > |T|$  of [1, Thm. 4.1] is used *only* in the applicability lemma [1, Lem. 4.2], which asserts that a usable run exists. The move itself [1, Lem. 4.3] requires only that a usable run *be given*, together with  $|T| \leq n - 2$ . So the object to compare with  $\mathcal{B}$  is not “the moves Theorem 4.1 guarantees” but “the moves Theorem 4.1 constructs”:

**Definition 1.1.** Let  $(S, T)$  be additive. A run  $[m, M]$  of  $S$  is *usable* if  $m \notin T$  and  $j + 1 \notin T$  for some  $j \in [m, M]$ . For a usable run put  $e_{[m,M]}$  = the first  $j \in \{m, m + 1, \dots, M\}$  with  $j + 1 \notin T$ . Set

$$\mathcal{E}(S, T) = \{e_{[m,M]} : [m, M] \text{ a usable run of } S\} \subseteq S,$$

and write  $\mathcal{E}^{\text{mv}}(S, T) = \{(S \setminus \{e\}, T \cup \{m\}) : e = e_{[m,M]}\}$  for the corresponding moves.

With this reading (a) is unrestricted after all, and (b) costs nothing: there is no regime in which  $\tilde{e}_1$  fires and Theorem 4.1 is silent. That is the content of Theorem 3.2. Once the comparison is set up this way, (Q) has a clean answer, and it is not the answer the question anticipated.

## 1.2 The answer

Write

$$\mathcal{B}(S, T) = \{b_x : x \in \mathcal{X}(S, T), \tilde{e}_1^{(x)}(uv) \neq \mathbf{0}\} \subseteq S, \quad b_x = \min_{\prec_x} L_1^{(x)}(uv),$$

so that  $\tilde{e}_1^{(x)}(uv)$  removes  $b_x$  from  $S$  and adds to  $T$  the bottom of the run of  $S$  containing  $b_x$ . Then:

- $\mathcal{B}(S, T) \subseteq \mathcal{E}(S, T)$  always, and the two moves agree letter for letter (Theorem 3.2). Every edge of every Morse–Schilling crystal on  $uv$ , at every admissible  $x$ , is an exchange move.
- The containment is strict. The smallest witnesses occur at  $n = 5$ ; at  $n = 5, 6, 7$  there are respectively 15, 132 and 854 exchange moves lying outside every Morse–Schilling crystal (Theorem 4.1 and §6).

- The difference is exactly the gap between local and global minimality of a height profile (Theorem 4.2).

So the sentence one would have liked — “Theorem 4.1 is the  $x$ -free shadow of the whole family of Morse–Schilling crystals, and dropping the pairing is taking the union over all of them” — is false. The true sentence is one-directional and, I think, more informative:

*The exchange move is strictly more general than the union of the entire family of Morse–Schilling crystal structures on  $uv$ . The  $x$ -pairing is what turns a local minimum into a global one; it buys a canonical, invertible operator and it costs coverage.*

**Remark 1.2.** Neither side of this comparison was built with the other in view, and neither has a free parameter I could tune. Morse–Schilling’s  $x$  is *their* parameter and I quantify over all of it; the exchange move has none. The implementation of [4, §3.2, Def. 3.2] used below reproduces all three of the worked examples printed in [4] exactly, for both the pairing and  $\tilde{e}_1$  (§6).

## 2 The two constructions

We use the block model of [1, §2]: for  $S \subsetneq \mathbb{Z}/n$ , identified with its  $n$ -periodic lift,  $u_S(j) = j - 1$  if  $j - 1 \in S$  and  $u_S(j) = \min\{c \geq j : c \notin S\}$  otherwise;  $\ell(u_S) = |S|$ . A *run* of  $S$  is a maximal interval  $[m, M] \subseteq \mathbb{Z}$  contained in  $S$ . Products act by  $(ab)(j) = a(b(j))$ , so  $u_S u_T$  means “ $u_T$  first”; this matches the factor order of [4], in which  $\tilde{e}_1$  moves a letter from the left factor to the right.

### 2.1 The Morse–Schilling pairing

Fix  $x \in \mathcal{X}(S, T)$ . Morse–Schilling order  $\mathbb{Z}/n \setminus \{x\}$  linearly by

$$x - 1 \succ x - 2 \succ \cdots \succ 0 \succ n - 1 \succ \cdots \succ x + 1, \quad (1)$$

and pair the  $\prec_x$ -largest  $b \in S$  with the  $\prec_x$ -smallest  $a \in T$  with  $a \succ_x b$ , ignoring already-paired letters, proceeding downwards through  $S$ .  $L_1(uv)$  and  $R_1(uv)$  are the unpaired elements of  $S$  and of  $T$ . Then [4, Def. 3.2(i)]

$$\tilde{e}_1(uv) = \tilde{u}\tilde{v}, \quad \text{content}(\tilde{u}) = S \setminus \{b\}, \quad \text{content}(\tilde{v}) = T \cup \{b - t\},$$

for  $b = \min_{\prec_x} L_1(uv)$  and  $t = \min\{i \geq 0 : b - i - 1 \notin S\}$ ; and  $\tilde{e}_1(uv) = \mathbf{0}$  if  $L_1(uv) = \emptyset$ . Note that  $b - t$  is precisely the bottom of the run of  $S$  containing  $b$ , which is the same letter Definition 1.1 adds to  $T$ . The two constructions therefore differ only in *which letter is removed from  $S$* , and the whole question is which elements of  $S$  each of them selects.

### 2.2 The word and its profile

**Definition 2.1.** Fix  $x \in \mathcal{X}(S, T)$  and let the *window* be the interval of integers  $(x, x + n] = \{x + 1, \dots, x + n\}$ , a lift of  $\mathbb{Z}/n$  with  $x$  itself omitted. Read the slots of the window in increasing order; at slot  $z$  emit the letter  $)$  if  $z \in T$ , and then the letter  $($  if  $z \in S$ . The resulting word is  $W_x$ . Define the *profile*  $P_x, Q_x$  on the window by

$$P_x(x) = 0, \quad Q_x(z) = P_x(z - 1) - [z \in T], \quad P_x(z) = Q_x(z) + [z \in S],$$

so  $Q_x(z)$  is the height of  $W_x$  at the low point of slot  $z$  and  $P_x(z)$  its height at the end of slot  $z$ .

Since  $P_x$  and  $Q_x$  obey a recursion independent of  $x$  and differ only in their initialisation, for two cuts  $x, x'$  and any  $z$  in both windows one has  $Q_x(z) - Q_{x'}(z) = \text{const}$ ; all statements below compare values of  $Q$  within one window and are therefore independent of the normalisation. Writing  $d = |S| - |T|$  we have  $Q(z+n) = Q(z) + d$ .

**Lemma 2.2** (The pairing is bracket matching). *For every  $x \in \mathcal{X}(S, T)$ , the  $uv$ -pairing with respect to  $x$  coincides with the standard bracket matching of  $W_x$ :  $b \in S$  is paired with  $a \in T$  if and only if the letter  $($  at slot  $b$  is closed by the letter  $)$  at slot  $a$ .*

*Proof.* Induction on  $|S|$ , the case  $S = \emptyset$  being vacuous. Let  $b_{\max}$  be the  $\prec_x$ -largest element of  $S$ , so the  $($  at slot  $b_{\max}$  is the last  $($  of  $W_x$ . If some  $)$  follows it, let  $a$  be the first such; there is no letter strictly between them, since any intervening letter would be a  $($ , contradicting maximality (within a slot,  $)$  precedes  $($ , so the  $)$  at slot  $a$  is the letter immediately after the  $($  at slot  $b_{\max}$  whenever  $a$  is the first  $)$  after it). Morse–Schilling process  $b_{\max}$  first and pair it with the  $\prec_x$ -smallest free  $a \succ_x b_{\max}$  in  $T$ , which is this  $a$ ; the bracket matching closes an adjacent  $()$ , which is this same pair. If no  $)$  follows  $b_{\max}$  then Morse–Schilling leave  $b_{\max}$  unpaired and the stack never pops it, so it is unmatched in the bracket matching too. In either case delete the matched pair (or the terminal  $()$  from  $W_x$ : deleting an adjacent  $()$  changes neither the bracket matching of the remaining letters nor the subsequent steps of the Morse–Schilling procedure, which continues with the next  $b$  and the remaining free letters of  $T$ . Apply the inductive hypothesis.  $\square$

**Lemma 2.3** (Unmatched letters read off the profile). *Let  $x \in \mathcal{X}(S, T)$  and  $z_0 \in S$  in the window. The letter  $($  at slot  $z_0$  is unmatched in  $W_x$  if and only if  $Q_x(z) > Q_x(z_0)$  for every  $z$  with  $z_0 < z \leq x+n$ .*

*Proof.* The  $($  at  $z_0$  is matched precisely when the height of  $W_x$  returns to its value  $Q_x(z_0)$  immediately before that letter at some later point of the word. The heights attained after that letter are  $P_x(z_0) = Q_x(z_0) + 1$  together with  $Q_x(z), P_x(z)$  for  $z_0 < z \leq x+n$ . Since  $P_x(z) \geq Q_x(z)$  for every  $z$ , the minimum of this collection equals  $\min\{Q_x(z) : z_0 < z \leq x+n\}$  when the window extends beyond  $z_0$ , and equals  $Q_x(z_0) + 1$  otherwise; in the latter case the stated condition is vacuously true and the letter is indeed unmatched. The claim follows.  $\square$

**Lemma 2.4** ( $b_x$  is the last global minimum of the profile). *Let  $x \in \mathcal{X}(S, T)$ , let  $\mu_x = \min\{Q_x(z) : x < z \leq x+n\}$ , and let  $z^*$  be the largest  $z$  in the window with  $Q_x(z) = \mu_x$ . Then  $L_1^{(x)}(uv) \neq \emptyset$  if and only if  $z^* \in S$ , and in that case  $b_x = z^*$ .*

*Proof.* Suppose  $z^* \in S$ . For every  $z > z^*$  in the window,  $Q_x(z) > \mu_x = Q_x(z^*)$  by maximality of  $z^*$ , so by Lemma 2.3 the letter at  $z^*$  is unmatched; hence  $L_1 \neq \emptyset$ . If  $z_0 \in S$  is unmatched and  $z_0 < z^*$  then Lemma 2.3 gives  $Q_x(z^*) > Q_x(z_0) \geq \mu_x = Q_x(z^*)$ , absurd; so  $z^*$  is the  $\prec_x$ -smallest unmatched letter, i.e.  $b_x = z^*$ .

Suppose instead  $z^* \notin S$ , and assume for contradiction that some  $z_0 \in S$  is unmatched. By the displayed argument  $z_0 \geq z^*$ , and  $z_0 \neq z^*$ , so  $z_0 > z^*$ . Let  $z_1$  be the smallest element of  $S$  in the window with  $z_1 > z^*$ ; then  $z_1 \leq z_0$  and every slot  $z$  with  $z^* < z < z_1$  lies outside  $S$ , so  $P_x(z) = Q_x(z)$  there, and  $P_x(z^*) = Q_x(z^*) = \mu_x$  as  $z^* \notin S$ . Since each slot can only decrease the height before its own low point,  $Q_x(z) \leq \mu_x$  for all  $z^* < z \leq z_1$ ; minimality of  $\mu_x$  forces equality, and in particular  $Q_x(z_1) = \mu_x$  with  $z_1 > z^*$  — contradicting the maximality of  $z^*$ . Hence no unmatched letter exists and  $L_1 = \emptyset$ .  $\square$

### 3 Every crystal edge is an exchange move

We use one lemma from [1], namely Step 1 of the proof of [1, Lem. 4.2]. It is the only place where length-additivity enters, and we restate it with its proof so that the present note is self-contained.

**Lemma 3.1** ([1, Lem. 4.2, Step 1]). *Let  $(S, T)$  be additive and let  $[m, M]$  be a run of  $S$  with  $m \in T$ . Then  $[m, M + 1] \subseteq T$ .*

*Proof.* Let  $[a, b]$  be the run of  $T$  containing  $m$ ; it is finite because  $T$  is proper. Apply the length-additivity criterion [1, Lem. 2.3] with  $w = u_S$  to the run  $[a, b]$  of  $T$ , legitimate since  $\ell(u_S u_T) = \ell(u_S) + |T|$ : it gives  $u_S(b + 1) > u_S(j)$  for all  $j \in [a, b]$ , in particular for  $j = m$ . Now  $m$  is the bottom of the  $S$ -block  $[m, M + 1]$ , so  $u_S(m) = M + 1$ . Suppose  $b \leq M$ . If  $b = m$  then  $b + 1 \in [m + 1, M + 1]$ , so  $u_S(b + 1) = m$ , and  $m > M + 1$  is false. If  $m < b \leq M$  then  $b + 1 \in [m + 1, M + 1]$ , so  $u_S(b + 1) = b \leq M < M + 1 = u_S(m)$ , again false. Hence  $b \geq M + 1$ , and since  $a \leq m$  we get  $[m, M + 1] \subseteq [a, b] \subseteq T$ .  $\square$

**Theorem 3.2.** *Let  $(S, T)$  be additive and let  $x \in \mathcal{X}(S, T)$  with  $\tilde{e}_1^{(x)}(u_S u_T) \neq \mathbf{0}$ . Put  $b = b_x$  and let  $[m, M]$  be the run of  $S$  containing  $b$ . Then*

- (i)  $b + 1 \notin T$ ;
- (ii)  $[m + 1, b] \subseteq T$ ;
- (iii)  $m \notin T$ .

Consequently  $[m, M]$  is a usable run,  $b = e_{[m, M]}$ , and

$$\tilde{e}_1^{(x)}(u_S u_T) = u_{S \setminus \{b\}} u_{T \cup \{m\}}$$

is exactly the Theorem 4.1 exchange move on the run  $[m, M]$ . In particular  $\mathcal{B}(S, T) \subseteq \mathcal{E}(S, T)$  and every  $\tilde{e}_1$ -image is a Theorem 4.1 move.

*Proof.* Throughout identify  $b$  and  $m$  with their lifts in the window  $(x, x + n]$ ; this is legitimate because  $[m, b] \subseteq S$  and  $x \notin S$ , so the whole interval  $[m, b]$  avoids the lift of  $x$  and lies in a single window, with  $m > x$ .

(i) First,  $b < x + n$ : indeed  $b \equiv x + n$  would give  $b \equiv x$ , impossible since  $b \in S$  and  $x \notin S$ . So the slot  $b + 1$  lies in the window. If  $b + 1 \in T$  then slot  $b + 1$  opens with a  $)$ , hence  $Q_x(b + 1) = P_x(b) - 1 = Q_x(b)$ , and Lemma 2.3 shows the letter at  $b$  is matched — contradicting  $b \in L_1^{(x)}$ .

(ii) Let  $m < j \leq b$ . Then  $j - 1 \in S$ , so  $P_x(j - 1) = Q_x(j - 1) + 1$  and hence  $Q_x(j) = Q_x(j - 1) + 1 - [j \in T]$ . Thus  $Q_x$  is non-decreasing on  $[m, b]$  and strictly increases exactly at those  $j$  with  $j \notin T$ . Suppose some  $j \in (m, b]$  has  $j \notin T$ . Then  $Q_x(j - 1) < Q_x(j) \leq Q_x(b)$ , and  $j - 1$  lies in the window; but  $b = b_x$  satisfies  $Q_x(b) = \mu_x$  by Lemma 2.4, so  $Q_x(j - 1) < \mu_x$  contradicts the definition of  $\mu_x$ . Hence every  $j \in (m, b]$  lies in  $T$ .

(iii) Suppose  $m \in T$ . By Lemma 3.1,  $[m, M + 1] \subseteq T$ . Since  $m \leq b \leq M$  we get  $b + 1 \in [m + 1, M + 1] \subseteq T$ , contradicting (i).

By (iii) and (i) the run  $[m, M]$  satisfies  $m \notin T$  and has an index, namely  $b$ , with  $b + 1 \notin T$ , so it is usable; by (ii) no  $j \in [m, b - 1]$  has  $j + 1 \notin T$ , so  $b$  is the *first* such index, i.e.  $b = e_{[m, M]}$ . Finally  $m \notin T$  and  $x \notin T$  with  $m \neq x$  give  $|T| \leq n - 2$ , so [1, Lem. 4.3] applies to the run  $[m, M]$  and its conclusion,  $u_{S \setminus \{b\}} u_{T \cup \{m\}} = u_S u_T$  with the factorisation length-additive, is the Theorem 4.1 move. Morse–Schilling’s operator removes  $b$  from  $S$  and adds  $b - t$ , the bottom of the run of  $S$  containing  $b$ , which is  $m$ ; the two moves coincide.  $\square$

**Remark 3.3.** The minimum in  $b = \min_{\prec_x} L_1$  is load-bearing, not a canonicalisation of an otherwise valid family. Part (ii) of the proof is the only step that uses minimality, and it fails for a general unmatched letter: for  $n = 3$ ,  $S = \{0, 1\}$ ,  $T = \emptyset$ ,  $x = 2$  the word is  $(($  and  $L_1 = \{0, 1\}$ , but only  $b = 0$  gives a factorisation — removing 1 and adding  $m = 0$  yields  $u_{\{0\}}u_{\{0\}} = s_0s_0 = \text{id}$ , of length 0, not 2. Over  $3 \leq n \leq 7$  the union  $\bigcup_x L_1^{(x)}$  fails to be contained in  $\mathcal{E}$  for 2752 of the 9093 additive pairs with  $\mathcal{X} \neq \emptyset$ .

## 4 The containment is strict, and where the excess lives

**Theorem 4.1.** *The containment of Theorem 3.2 is strict. The smallest witnesses occur at  $n = 5$ ; there are no witnesses for  $n \leq 4$ .*

*Proof.* Take  $n = 5$ ,  $S = \{0, 2\}$ ,  $T = \{4\}$ , so  $u_S = s_0s_2$ ,  $u_T = s_4$  and  $v = s_0s_2s_4$  has length  $3 = |S| + |T|$ : the pair is additive. Note  $|S| = 2 > 1 = |T|$ , so the hypothesis of [1, Thm. 4.1] holds and  $\tilde{e}_1^{(x)}$  fires at every admissible  $x$ .

*The exchange moves.*  $S$  has the two runs  $[0, 0]$  and  $[2, 2]$ . For  $[0, 0]$ :  $0 \notin T$  and  $1 \notin T$ , so the run is usable with  $e = 0$ , giving  $u_{\{2\}}u_{\{0,4\}} = s_2 \cdot s_0s_4 = s_0s_2s_4 = v$ . For  $[2, 2]$ :  $2 \notin T$  and  $3 \notin T$ , so the run is usable with  $e = 2$ , giving  $u_{\{0\}}u_{\{2,4\}} = s_0 \cdot s_2s_4 = v$ . Both are length-additive. Hence  $\mathcal{E}(S, T) = \{0, 2\}$ .

*The crystal edges.*  $\mathcal{X}(S, T) = \{1, 3\}$ . For  $x = 1$  the window is  $\{2, 3, 4, 5\} \cup \{6\}$ , whose word is  $(($  at slots 2, 4, 5; the letters at 2 and 4 match and the letter at  $5 \equiv 0$  is unmatched, so  $L_1 = \{0\}$  and  $b_1 = 0$ . For  $x = 3$  the window is  $\{4, \dots, 8\}$ , whose word is  $)(($  at slots 4, 5, 7; here  $L_1 = \{0, 2\}$ , and since (1) for  $x = 3$  reads  $2 \succ 1 \succ 0 \succ 4$ , the  $\prec_3$ -minimum is  $b_3 = 0$ . Hence  $\mathcal{B}(S, T) = \{0\} \subsetneq \{0, 2\} = \mathcal{E}(S, T)$ : the exchange move on the run  $[2, 2]$  is an edge of no Morse–Schilling crystal.

The mechanism is visible in the profile. Normalising  $P(-1) = 0$  one finds  $Q(0) = 0$ ,  $Q(1) = 1$ ,  $Q(2) = 1$ ,  $Q(3) = 2$ ,  $Q(4) = 1$  and  $d = |S| - |T| = 1$ , so  $Q(5) = Q(0) + 1 = 1 = Q(2)$ . In every window of length  $n$  containing both, the slots 2 and 5 carry equal profile values and slot  $5 \equiv 0$  comes later; by Lemma 2.4 the operator selects the *last* minimiser, so slot 2 is a minimiser that is never the last one. Exhaustive verification for  $n \leq 4$  (§6) shows  $\mathcal{B} = \mathcal{E}$  there.  $\square$

**Theorem 4.2** (Local versus global minimality). *Let  $(S, T)$  be additive with  $\mathcal{X}(S, T) \neq \emptyset$  and let  $c \in S$ . Then:*

1.  *$c \in \mathcal{E}(S, T)$  if and only if, with  $m$  the bottom of the run of  $S$  containing  $c$ , one has  $m \notin T$ ,  $[m + 1, c] \subseteq T$  and  $c + 1 \notin T$ . Equivalently:  $Q$  is constant on  $[m, c]$ ,  $Q$  increases strictly at  $c + 1$ , and  $m \notin T$ . This is a condition on the single run of  $c$  — it is local.*
2.  *$c \in \mathcal{B}(S, T)$  if and only if  $c \in \mathcal{E}(S, T)$  and there exists  $x \in \mathcal{X}(S, T)$ , lifted to  $\hat{x}$  with  $c - n \leq \hat{x} < c$ , such that*

$$Q(z) \geq Q(c) \quad (\hat{x} < z < c) \quad \text{and} \quad Q(z) > Q(c) \quad (c < z \leq \hat{x} + n).$$

*Equivalently, writing*

$$F(c) = \min\{z > c : Q(z) \leq Q(c)\}, \quad G(c) = \max\{z < c : Q(z) < Q(c)\}$$

*(with  $F = +\infty$ ,  $G = -\infty$  if no such  $z$  exists),  $c \in \mathcal{B}(S, T)$  if and only if the integer interval  $[\max(G(c), c - n), \min(F(c) - n - 1, c - 1)]$  contains a lift of an element of  $\mathcal{X}(S, T)$ . This is a condition on a full period — it is global.*

*Proof.* (1) is Definition 1.1 unwound:  $[m, M]$  is usable with  $e_{[m, M]} = c$  exactly when  $m \notin T$ ,  $j + 1 \in T$  for  $m \leq j < c$  and  $c + 1 \notin T$ ; the first of these two conditions is  $[m + 1, c] \subseteq T$ . The reformulation in terms of  $Q$  is the computation already made in the proof of Theorem 3.2(ii): for  $m < j \leq c$  one has  $Q(j) = Q(j - 1) + 1 - [j \in T]$ , so  $[m + 1, c] \subseteq T$  says precisely that  $Q$  is constant on  $[m, c]$ , and  $c + 1 \notin T$  says  $Q(c + 1) = P(c) = Q(c) + 1$ .

(2) By Lemma 2.4, for  $x \in \mathcal{X}(S, T)$  we have  $b_x = c$  if and only if  $c$  lies in the window  $(x, x + n]$  and is the largest slot of that window at which  $Q_x$  attains its minimum over the window. Since  $Q_x$  and  $Q$  differ by a constant on the window, this says exactly  $Q(z) \geq Q(c)$  for  $\hat{x} < z < c$  and  $Q(z) > Q(c)$  for  $c < z \leq \hat{x} + n$ , which is the displayed condition. The first clause holds for all  $z \in (\hat{x}, c)$  iff  $\hat{x} \geq G(c)$ , and the second holds for all  $z \in (c, \hat{x} + n]$  iff  $\hat{x} + n < F(c)$ , i.e.  $\hat{x} \leq F(c) - n - 1$ ; together with  $c - n \leq \hat{x} < c$  this is the stated interval. Finally,  $b_x = c$  forces  $c \in \mathcal{E}(S, T)$  by Theorem 3.2.  $\square$

**Remark 4.3.** This is what the pairing buys. A crystal operator must be a *function*: it must select one of the available moves, and it must do so in a way compatible with  $\tilde{f}_1$ , with the Stembridge axioms, and with the weight. Morse–Schilling’s selection rule is “take the last global minimum of the profile over the window”, and globality is precisely what the  $x$ -pairing computes — a pairing is a statement about the whole word, a run is not. Theorem 4.1 asks only for local minimality and therefore sees moves that no global rule can reach: in the example of Theorem 4.1, slot 2 ties with a later slot in every window, and a rule that must break ties consistently can never choose it. Conversely, having no parameter, the exchange move makes no claim to canonicity:  $\mathcal{E}(S, T)$  is a set, not an operator, and Theorem 4.1 asserts only that it is nonempty when  $|S| > |T|$ , which is all [1, Cor. 4.5] consumes.

## 5 The case $\mathcal{X}(S, T) = \emptyset$

When  $S \cup T = \mathbb{Z}/n$  there is no admissible  $x$  and Definition 2.1 does not apply. Morse–Schilling handle this in [4, §3.4]: they bracket  $i \in S$  with  $i + 1 \in T$ , show (their existence lemma for Case-(3) indices, Lemma 3.13) that some index is in “Case (3)”, i.e. lies in a maximal block

$$\{b - 1, \dots, b - t\} \subseteq S, \quad b \notin S, \quad \{b - t + 1, \dots, b\} \subseteq T, \quad b - t \notin T,$$

and then remove the bracketed pair  $(b - 1, b)$  and proceed with  $x = b$ .

Two points of honesty. First, the prescription “proceed with  $x = b$ ” leaves it unstated whether the offset  $t$  of [4, Def. 3.2(i)] is computed against  $\text{content}(u)$  or against the reduced content  $\text{content}(u) \setminus \{b - 1\}$ ; we implemented both readings. Second, the choice of Case-(3) index  $b$  is not in general unique, so this construction, like the main one, defines a *family* of operators, and we take the union over the family. With those two conventions fixed, exhaustive computation for  $3 \leq n \leq 7$  gives, over all 1636 additive pairs with  $\mathcal{X} = \emptyset$ :

- both readings produce the same set of moves in every case;
- every move so produced is a genuine length-additive factorisation of  $us_{UT}$  (1636 of 1636 pairs, checked by comparing window notations directly);
- every move so produced lies in  $\mathcal{E}^{\text{mv}}(S, T)$  (1636 of 1636), and the containment is again strict, failing to be an equality in 7 pairs, all at  $n = 7$ .

So Theorem 3.2 and Theorem 4.1 are not artefacts of restricting to  $\mathcal{X} \neq \emptyset$ : the conclusion is uniform over all two-factor affine factorisations. We record the  $\mathcal{X} = \emptyset$  statement as computational, not proved — see §7.

## 6 Computational verification

All code is in `proofs/code-q220/` of the commit named on the title page. Affine permutations are held in window notation and lengths computed by Shi’s inversion formula, not by the block model, so the block model of [1, Lem. 2.2] is tested rather than assumed.

**Instrument validation.** The implementation of [4, §3.2 (the  $uv$ -pairing)] reproduces all three pairings printed in [4, Ex. 3.1] exactly, including the pair labels ( $n = 14$ ,  $x = 10$ : pairs  $(5, 6), (2, 4), (12, 13)$ ,  $L_1 = \{8, 9\}$ ;  $n = 14$ ,  $x = 3$ : pairs  $(12, 13), (9, 11), (8, 0), (5, 6)$ ;  $n = 5$ ,  $x = 2$ : pair  $(0, 1)$ ), and the implementation of [4, Def. 3.2(i)] reproduces all three  $\tilde{e}_1$ -images printed in [4, Ex. 3.4] exactly. No parameter was fitted.

**The lemmas.** Lemma 2.2 was checked on every admissible cut of every additive pair for  $3 \leq n \leq 7$  (17,368 cuts, 0 disagreements between the Morse–Schilling procedure and bracket matching). Lemma 2.4 was checked on the same 17,368 cuts, including the “only if” half (0 failures). Lemma 3.1 was checked on all 3,742 instances of its hypothesis (0 failures).

**Theorem 3.2.** Clauses (i), (ii), (iii), the identification  $b = e_{[m, M]}$ , and the equality of the two moves were each checked separately at all 10,877 firings of  $\tilde{e}_1^{(x)}$  over  $3 \leq n \leq 7$ : 0 failures in each of the five tests. Independently, all 9,120 Theorem 4.1 moves were re-checked to be genuine length-additive factorisations with proper supports (0 failures); this tests the implementation of [1, Thm. 4.1], not the theorem.

**Theorem 4.2.** The criterion of part (2) was evaluated independently of the pairing, from the profile alone, and compared with  $\mathcal{B}(S, T)$  computed from [4, Def. 3.2]: agreement on 9,093 of 9,093 additive pairs with  $\mathcal{X} \neq \emptyset$ .

**The counts.** For  $3 \leq n \leq 7$ :

| $n$ | additive pairs | $\mathcal{X} = \emptyset$ | $ \mathcal{E} $ | $ \mathcal{B} $ | pairs with $\mathcal{B} = \mathcal{E}$ | excess letters |
|-----|----------------|---------------------------|-----------------|-----------------|----------------------------------------|----------------|
| 3   | 34             | 9                         | 15              | 15              | 34                                     | 0              |
| 4   | 143            | 34                        | 80              | 80              | 143                                    | 0              |
| 5   | 561            | 115                       | 375             | 360             | 546                                    | 15             |
| 6   | 2122           | 365                       | 1650            | 1518            | 1990                                   | 132            |
| 7   | 7869           | 1113                      | 7000            | 6146            | 7015                                   | 854            |

Here  $|\mathcal{E}|$  and  $|\mathcal{B}|$  are summed over all additive pairs,  $\mathcal{B}$  being computed with the  $\mathcal{X} = \emptyset$  convention of §5 where needed, and “excess letters” counts  $\sum |\mathcal{E} \setminus \mathcal{B}|$ .  $\mathcal{B} \subseteq \mathcal{E}$  held in all 10,729 pairs.

**Remark 6.1.** No coverage *fraction* is quoted anywhere in this note. Two independent implementations of the Morse–Schilling pairing agreed on the forward direction on 21 September and disagreed on a converse coverage fraction; that discrepancy is unresolved, so only integer counts, each with its defining predicate spelled out, appear above.

## 7 What this consumes, and what remains open

**Consumed.** §3.2, Definition 3.2 and §3.4 of [4], read at source; Lemmas 2.2, 2.3, 4.2 (Step 1) and 4.3 and Theorem 4.1 of [1]. Nothing is consumed from [3], from the theory of dual  $k$ -Schur

functions, or from [4, Thm. 3.5] (that  $B(w)$  is a crystal): Theorem 3.2 is a statement about the operator  $\tilde{e}_1$ , not about the crystal axioms it satisfies.

### Gaps.

- (G1) §5 is *computational, not proved*. The obstruction is precise: Theorem 3.2(iii) invokes Lemma 3.1, which needs  $(S, T)$  additive, whereas the  $\mathcal{X} = \emptyset$  construction pairs the *reduced* pair  $(S \setminus \{b-1\}, T \setminus \{b\})$ , which need not be additive. I do not know whether Theorem 3.2 holds there for every choice of Case-(3) index; it holds in all 1636 instances with  $n \leq 7$ .
- (G2) The reading of “ $t$  maximal” and of “proceed with  $x = b$ ” in [4, §3.4] is mine, not theirs. Both readings I could construct give the same answer, which is evidence that the ambiguity is immaterial, not proof.
- (G3) Theorem 4.2(2) characterises  $\mathcal{B}$  by a condition that still quantifies over  $\mathcal{X}$ . A characterisation intrinsic to  $(S, T)$  — one that predicts  $|\mathcal{E} \setminus \mathcal{B}|$  from the run structure without sliding a window — would be better, and I do not have one. The sequence of excess counts 0, 0, 15, 132, 854 for  $n = 3, \dots, 7$  is offered as data, not as a conjecture.
- (G4) Theorem 3.2 says every  $\tilde{e}_1$ -edge is an exchange move; it says nothing about  $\tilde{f}_1$  or  $\tilde{s}_1$ . Since  $\tilde{f}_1$  is inverse to  $\tilde{e}_1$  where both are defined, one expects the corresponding statement for  $\tilde{f}_1$  to follow from the dual exchange move obtained in the proof of [1, Cor. 4.5] by conjugating with the length-preserving anti-automorphism  $\Psi$ ; I have not checked this and do not claim it.

## References

- [1] Clio, *An exchange move for cyclically decreasing factorisations, and the reduction of WZZ Problem 5.1 to a single hypothesis*, 21 September 2026, [clio-vega/proofs@22d1b22](#).
- [2] T. Lam, *Affine Stanley symmetric functions*, Amer. J. Math. 128 (2006), 1553–1586.
- [3] T. Lam, *Schubert polynomials for the affine Grassmannian*, J. Amer. Math. Soc. 21 (2008), 259–281.
- [4] J. Morse and A. Schilling, *Crystal approach to affine Schubert calculus*, [arXiv:1408.0320](#).